

Attention Is All You Need: A Comprehensive Study

Ashish Vaswani, Noam Shazeer, Niki Parmar, et al.

Abstract

Positional encoding uses sine and cosine functions: $PE(pos, 2i) = \sin(pos/10000^{(2i/d_model)})$, $PE(pos, 2i+1) = \cos(pos/10000^{(2i/d_model)})$. This allows learning relative positions since $PE(pos+k)$ is a linear function of $PE(pos)$. Learned embeddings performed equally well. The base model has $d_model=512$, 6 encoder layers, 6 decoder layers, 8 heads, $d_ff=2048$. The large model has $d_model=1024$, 12 layers, 16 heads, $d_ff=4096$. Training uses Adam with $lr=d_model^{(-0.5)} \cdot \min(step^{(-0.5)}, step \cdot warmup_steps^{(-1.5)})$, $warmup_steps=4000$. The decoder has $N=6$ layers with three sub-layers: masked multi-head self-attention, multi-head attention over encoder output, and position-wise FFN. Masking ensures predictions for position i depend only on known outputs at positions less than i . Attention uses $d_k=d_v=d_model/h=64$ with $h=8$ parallel heads. The encoder maps input symbols to continuous representations via $N=6$ identical layers, each with two sub-layers: multi-head self-attention and position-wise feed-forward network. Residual connections around each sub-layer are followed by layer normalization. The FFN has two linear transforms with ReLU, inner dimension $d_ff=2048$.

1 Introduction

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The Transformer relies entirely on self-attention to compute representations without recurrence or convolution. Multi-head attention allows jointly attending to information from different representation subspaces at different positions. Each head computes scaled dot-product attention on queries, keys, and values from learned linear projections. Positional encoding uses sine and cosine functions: $\text{PE}(\text{pos}, 2i) = \sin(\text{pos}/10000^{(2i/d_{\text{model}})})$, $\text{PE}(\text{pos}, 2i+1) = \cos(\text{pos}/10000^{(2i/d_{\text{model}})})$. This allows learning relative positions since $\text{PE}(\text{pos}+k)$ is a linear function of $\text{PE}(\text{pos})$. Learned embeddings performed equally well. The encoder maps input symbols to continuous representations via $N=6$ identical layers, each with two sub-layers: multi-head self-attention and position-wise feed-forward network. Residual connections around each sub-layer are followed by layer normalization. The FFN has two linear transforms with ReLU, inner dimension $d_{\text{ff}}=2048$.

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2 Related Work

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3 Method

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Table 1: Benchmark comparison

Method	Dataset	Acc (%)	F1
Baseline	ImageNet	72.3	0.71
ResNet-50	ImageNet	76.1	0.75
ResNet-152	ImageNet	78.3	0.77
Ours	ImageNet	79.1	0.78

4.1 Datasets and Setup

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4.2 Main Results

The Transformer relies entirely on self-attention to compute representations without recurrence or convolution. Multi-head attention allows jointly attending to information from different representation subspaces at different positions. Each head computes scaled dot-product attention on queries, keys, and values from learned linear projections. The encoder maps input symbols to continuous representations via $N=6$ identical layers, each with two sub-layers: multi-head self-attention and position-wise feed-forward network. Residual connections around each sub-layer are followed by layer normalization. The FFN has two linear transforms with ReLU, inner dimension $d_{\text{ff}}=2048$. The decoder has $N=6$ layers with three sub-layers: masked multi-head self-attention, multi-head attention over encoder output, and position-wise FFN. Masking ensures predictions for position i depend only on known outputs at positions less than i . Attention uses $d_k=d_v=d_{\text{model}}/h=64$ with $h=8$ parallel heads.

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On WMT 2014 En-De, the big model achieves 28.4 BLEU on newstest2014, outperforming ensembles by over 2 BLEU. On WMT 2014 En-Fr, it achieves 41.8 BLEU after 3.5 days on 8 GPUs, a new single-model SOTA. Label smoothing with $\epsilon=0.1$ improves both BLEU and accuracy. Positional encoding uses sine and cosine functions: $\text{PE}(\text{pos}, 2i) = \sin(\text{pos}/10000^{(2i/d_{\text{model}})})$, $\text{PE}(\text{pos}, 2i+1) = \cos(\text{pos}/10000^{(2i/d_{\text{model}})})$. This allows learning relative positions since $\text{PE}(\text{pos}+k)$ is a linear function of $\text{PE}(\text{pos})$. Learned embeddings performed equally well. The base model has $d_{\text{model}}=512$, 6 encoder layers, 6 decoder layers, 8 heads, $d_{\text{ff}}=2048$. The large model has $d_{\text{model}}=1024$, 12 layers, 16 heads, $d_{\text{ff}}=4096$. Training uses Adam with $\text{lr}=d_{\text{model}}^{-(0.5)} \cdot \min(\text{step}^{-(0.5)}, \text{step} \cdot \text{warmup_steps}^{-(1.5)})$, $\text{warmup_steps}=4000$.

Table 1: Benchmark comparison

Method	Dataset	Acc (%)	F1
Baseline	ImageNet	72.3	0.71
ResNet-50	ImageNet	76.1	0.75
ResNet-152	ImageNet	78.3	0.77
Ours	ImageNet	79.1	0.78

5 Analysis

Positional encoding uses sine and cosine functions:

$PE(pos, 2i) = \sin(pos/10000^{(2i/d_model)})$, $PE(pos, 2i+1) = \cos(pos/10000^{(2i/d_model)})$.

This allows learning relative positions since $PE(pos+k)$ is a linear function of $PE(pos)$. Learned embeddings performed equally well. The decoder has $N=6$ layers with three sub-layers: masked multi-head self-attention, multi-head attention over encoder output, and position-wise FFN. Masking ensures predictions for position i depend only on known outputs at positions less than i . Attention uses $d_k=d_v=d_model/h=64$ with $h=8$ parallel heads. The Transformer relies entirely on self-attention to compute representations without recurrence or convolution. Multi-head attention allows jointly attending to information from different representation subspaces at different positions. Each head computes scaled dot-product attention on queries, keys, and values from learned linear projections.

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